

GAIA-103 · Sonnet 4.6

- R10 WikiTextFetch ships;
+4.9pp (5/7 flips)
- R11 Trace: subset passes via
format exploit(verbatimim
match, no cross-check)
- R12 Guard: require 2nd
retrieval path



Genuine retrieval gain and verifier exploit co-ship at R10; trace catches exploit at R11; guard removes hacking path at R12.

WebShop · Sonnet 4.6

- R1 Anti-loop ship, SR
+16pp, reward +0.16
- R5 Reward up, SR flat;
gap +0.065 (overlap
OK, exact-match)
- R7 Decoupling peaks;
gap +0.113 (reward
0.87 vs SR 0.76)



Near-match buys farm partial reward: mean-reward climbs while exact-match SR stays frozen at 76% (gap +0.113).

SWE-bench · Qwen

- R5 All gates pass; score
20 → 16 (form OK,
capability ↓)
- R9 Gates pass again; 19
→ 14 (same pattern)
- R12 Contrast: workspace
ship → real +3 (only
structural win)



Gates check replay/novelty/structure
→ not pass rate; compliant ships can regress.

(a) Reward hacking (GAIA,
Sonnet 4.6).

(b) Reward hacking
(WebShop, Sonnet 4.6).

(c) Reward hacking (SWE-
bench, GPT-5.4).

τ³-Bench Tel · Sonnet

- R2~6 5 same-bucket edits
→ peak 100% (R4)
- R7 6th edit, rules couple:
94.7 → 80.7 (-14.0pp)
- R8~9 Ensemble recovers →
99.1%



Critic flagged bucket concentration but shipped:pass@2 misses coupling. Dip transient, not permanent.

SWE-bench · Sonnet

- R2~3 Climb to peak ~61.8%
(35/57 scale)
- R4~5 Post-peak drop to
~50.9% (prompt ships)
- R9 Verification Gate ships;
breaks 2 tasks with
9-round P-streaks
- R10 Phantom-regression
chase: recover 3



Most dips recover next round; CF is moderate churn, not permanent wipe-out.

GAIA-103 · GPT-5.4

- R1~4 Advisory only →
peak 73.8% (R4)
- R5~6 Enforcement cliff:
SearchBudget +
ComputationGate, 76
→ 55 (-21)
- R12 Rollback fails; final
49.5% (-24.3pp)



Each enforcement edit passes gate alone;
sub-threshold accumulation erases R4 gains.

(d) Forgetting (τ³-Bench
Telecom, Sonnet 4.6).

(e) Forgetting (SWE-bench,
GPT-5.4).

(f) Forgetting (GAIA,
GPT-5.4).

ALFWorld · Sonnet 4.6

- R4~5 Prompt-only edits,
gains stall; < 1pp/R
- R6 Lone processor edit;
14% (1/7), reverted
- R7~8 Ship-prediction → 0%,
then no_op; plateau
94.8%



Prompt space exhausted by R4; the lone structural edit flopped (14%) and reverted; ship-prediction 80% → 0%.

GAIA-103 · Sonnet 4.6

- R1~9 0 tools ships;
Blocked-source 39%
(prompt/config only)
- R9 Critic flags tools neglect
and still no ship
- R10 WikiTextFetch;
+4.9pp, hit rate 0.71



Trace pointed at tools from R1; pipeline stayed local until R10 structural tool ship.

WebShop · Sonnet 4.6

- R1~7 0 tools ships; prompt-
locked (hit-rate 0.31)
- R7 Critic flags tools bucket
untouched; still no ship
- R8 First tool ship
(ColorMatcher); -4pp
(76 → 72), 2/5



Pipeline stayed prompt-local for 7
rounds; the belated R8 tool ship regressed

(g) Under-exploration
(ALFWorld, Sonnet 4.6).

(h) Under-exploration
(GAIA, GPT-5.4).

(i) Under-exploration
(WebShop, Sonnet 4.6).